A STRONGLY REGULAR GRAPH WITH PARAMETERS $(99, 14, 1, 2)$ HAS NO AUTOMORPHISM OF ORDER TWO

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Abstract. Conway's 99-graph problem asks whether there is a strongly regular graph with parameters $(99, 14, 1, 2)$. We prove that such a graph has no automorphism of order 2. The first half of the proof is done by hand: from the definition of the graph, an involution produces a tight frame of 51 vectors inside an odd unimodular lattice of rank 24 without vectors of norm 1, hence inside one of the 156 lattices of Borcherds' classification. The second half is computer-assisted and shows that none of the 156 lattices contains such a frame. Twenty lattices are excluded by short arguments on the lattice itself. For the 136 others, the pairs of vectors of norm 7 that could be the two normals of the frame are enumerated up to symmetry and excluded one by one: by a short vector whose existence contradicts one of three nonexistence lemmas for integer eigenfunctions of small norm, with an exact certificate for each pair, or by an exhaustive search for the frame. We state which steps were verified by independent programs and which rest on a single audited program. All programs, records and certificates are available in the repository of the paper. Together with the known exclusions of the odd prime orders, the theorem implies that a strongly regular graph with parameters $(99, 14, 1, 2)$, if it exists, has no nontrivial automorphism; that consequence and its dependencies are the subject of a separate note.

1. Introduction

A strongly regular graph with parameters (v, k, λ, μ) is a k-regular graph on v vertices in which two adjacent vertices have exactly λ common neighbours and two distinct non-adjacent vertices exactly μ. Whether a strongly regular graph with parameters $(99, 14, 1, 2)$ exists is one of Conway's five problems [6]. Its adjacency matrix would have the eigenvalues 14, 3 and -4 with multiplicities 1, 54 and 44, and the standard feasibility conditions do not exclude it. The parameters belong to the family $(v, k, 1, 2)$, in which k is one of 4, 14, 22, 112, 994 [17]. The graphs with $k = 4$ (the 3×3 rook's graph, on 9 vertices) and with $k = 22$ (on 243 vertices, from the ternary Golay code) exist; to our knowledge the three other cases are open. The case $k = 14$ was first studied by Wilbrink [20]; we refer to [3] for strongly regular graphs in general.

Most of what is known concerns the automorphism group G of such a graph Γ. Makhnev and Minakova [17] proved by character theory that $|G|$ divides $2 \cdot 3^3 \cdot 7 \cdot 11$; if $|G|$ is even it divides 42 [17, 16]. Behbahani and Lam [1, 2] showed, by a computer enumeration of orbit matrices, that the only primes that can divide $|G|$ are 2 and 3, and that an automorphism of order 3 has no fixed point. Crnković and Maksimović [9] excluded, also by computer, groups of automorphisms of order 6 and 9. Cesarz and Woldar [5] proved without a computer that $G \cong \mathbb{Z}_7$ if 7 divides $|G|$, and consequently that $|G|$ divides 6 if it is even; this does not exclude an automorphism of order 7, which remained open without a computer. Together these results left $|G| \in \{1, 2, 3\}$. Two manuscripts of the present project treat odd orders: order 3 is excluded

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in [12] by linear algebra over $\mathbb{Z}$ and the Eisenstein integers, and order 7 is excluded without a computer in [13], starting from the orbit structure and the counting identities of [5]; both come with a formalisation in Lean 4. This paper treats order 2.

Theorem 1.1. *A strongly regular graph with parameters* $(99, 14, 1, 2)$ *has no automorphism of order* 2*.*

The proof of Theorem 1.1 uses none of the results quoted above. Combined with them it gives the following statement, whose dependencies are examined one by one in the separate note [14].

Corollary 1.2. *A strongly regular graph with parameters* $(99, 14, 1, 2)$*, if it exists, has no nontrivial automorphism.*

Proof. A nontrivial finite group has an element of prime order p, and $p \in \{2, 3, 7, 11\}$ by [17]. Order 2 is excluded by Theorem 1.1, order 3 by [12], and orders 7 and 11 by the computation of [2] (order 7 also without a computer, by [13]). □

Outline. Let σ be an automorphism of order 2 of Γ. A counting argument shows that σ fixes exactly one vertex (Section 3); this is known from the study of fixed-point subgraphs in [17], and we include a short direct proof. On the 49-dimensional space of σ-odd functions the adjacency matrix acts by an integer matrix S with $S^2 + S = 12I$, and $H = 3I - S$ is the Gram matrix of 49 vectors forming a tight frame of rank 22 (Section 4). The lattice N they generate has determinant a power of 7. The oddity formula determines the Witt class of its discriminant form, and a double count shows that the dual lattice N^* has minimum at least $8/7$ (Section 5). Gluing an overlattice of N with two orthogonal vectors of norm 7 gives an odd unimodular lattice U of rank 24 without vectors of norm 1 (Theorem 6.1); there are 156 such lattices [4].

The second half excludes the frame from each of the 156 lattices. Every vector v of U outside the plane of the two vectors of norm 7 yields an integer-valued σ-odd eigenfunction of the adjacency matrix for the eigenvalue -4, whose squared norm is $2E(v)$ for an explicit quadratic "energy" E. Three lemmas state that such eigenfunctions do not exist for the squared norms 16, 18 and 20 (Section 7); with the bound $8/7$ this forbids every such vector of energy less than 11. For a large part of the candidate pairs (y_1, y_2) of normals this is already a contradiction; the others are excluded by an exhaustive search for the frame (Section 8). Section 9 says how the computations were verified and what the proof rests on. In the way computations enter the proof we have tried to follow the practice of the nonexistence proof of a projective plane of order 10 [15]: the algorithm is described, the exhaustiveness of each search is argued in the text, and the consistency checks and independent recomputations are listed with their exact scope.

2. Notation

Throughout, Γ is a strongly regular graph with parameters $(99, 14, 1, 2)$, A its adjacency matrix, J the all-ones matrix and $N(x)$ the set of neighbours of a vertex x. Thus

$$A^2 + A - 12I = 2J. \tag{1}$$

Since $\lambda = 1$, the neighbourhood $N(x)$ induces seven disjoint edges, which we call the *blocks* at x; every edge lies in exactly one triangle. A *lattice* is a free $\mathbb{Z}$-module of finite rank with a positive definite rational symmetric bilinear form $(\ ,\ )$; the *norm* of v is (v, v); L^* denotes the dual lattice. A *root* is a vector of norm 2.

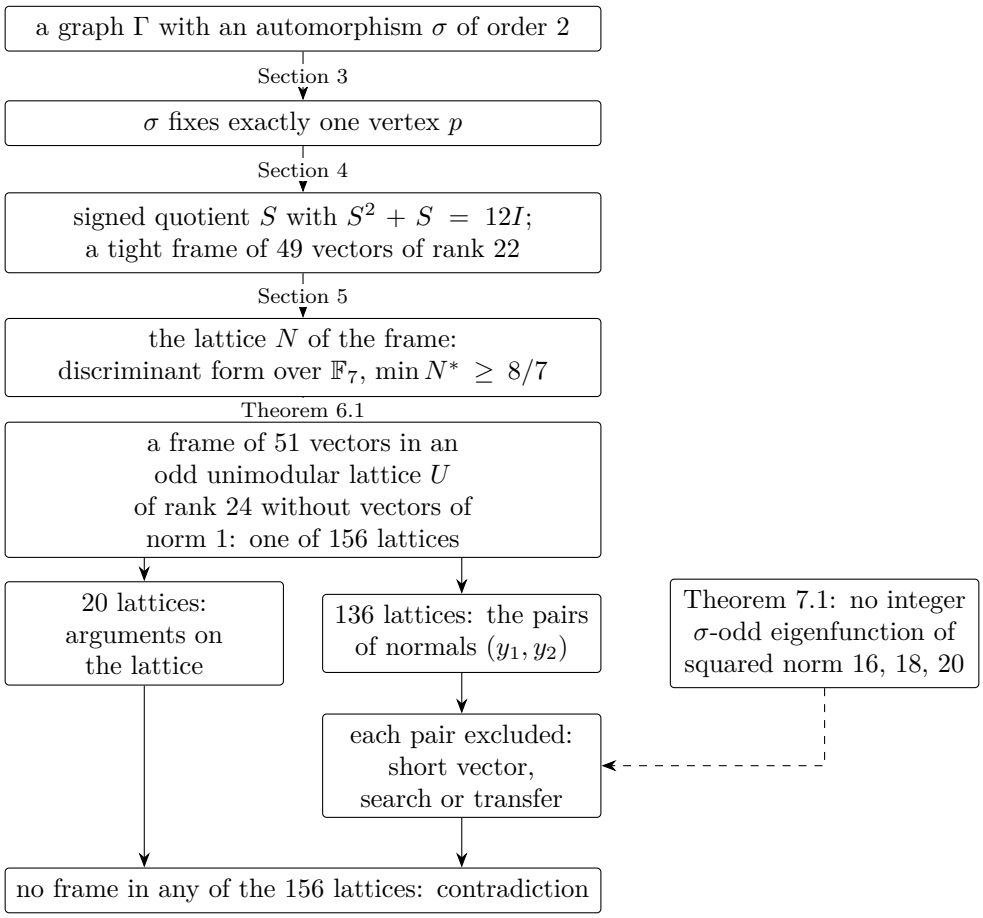


FIGURE 1. The chain of the proof. The left column is proved by hand in Sections 3–6; the exclusion of the 156 lattices (Section 8) and the lemmas of Theorem 7.1 are computer-assisted.

3. An involution fixes exactly one vertex

Proposition 3.1. *Let σ be an automorphism of order 2 of Γ. Then σ fixes exactly one vertex p, it exchanges the two ends of each of the seven blocks at p, and no vertex outside $N(p)$ is adjacent to its image.*

Proof. Let F be the set of fixed vertices and $f = |F|$; $f \geq 1$ because 99 is odd. Let $p \in F$. Then σ permutes the seven blocks at p. A block with a fixed end is fixed pointwise, the other end being the unique common neighbour of p and the first. Let s_p be the number of blocks fixed pointwise, t_p the number of blocks whose ends are exchanged, and $2r_p$ the number of blocks sent to another block, so that

$$s_p + t_p + 2r_p = 7. \tag{2}$$

A vertex z at distance 2 from p is determined by its two neighbours a, b in $N(p)$, which lie in different blocks, and all 84 such pairs occur. The vertex z is fixed if and only if $\{a, b\}$ is σ-invariant: either a, b are both fixed ($2s_p(s_p - 1)$ pairs) or $b = \sigma a$ lies in another block ($2r_p$ pairs). Counting p and its $2s_p$ fixed neighbours,

$$f = 1 + 2s_p^2 + 2r_p \qquad \text{for every } p \in F. \tag{3}$$

Let a_1 be the number of vertices v with $v \sim \sigma v$. For such v the unique common neighbour of v and σv is a fixed vertex p, and $\{v, \sigma v\}$ is an exchanged block at p; hence $a_1 = 2\sum_{p\in F} t_p$. A vertex v in a moved block at p is not adjacent to σv; the common neighbours of v and σv are p and one further vertex, which is fixed; and every fixed neighbour of v is a common neighbour of v and σv. So v has exactly two fixed neighbours, and the number of such vertices is $n_2 = 2\sum_{p\in F} r_p$. These vertices are not fixed and are distinct from those counted by a_1:

$$a_1 + n_2 \le 99 - f. \tag{4}$$

Each triangle fixed pointwise contributes to s_p for its three vertices:

$$\textstyle\sum_{p\in F} s_p \equiv 0 \pmod 3. \tag{5}$$

Finally, the idempotent of the eigenvalue -4 is $\frac{1}{7}(3I - A) + \frac{1}{63}J$, and the trace of σ on that eigenspace is an integer:

$$\tfrac{1}{7}\,(3f - a_1 + 11) \in \mathbb{Z}. \tag{6}$$

By (2) and (3) the triple (s_p, r_p, t_p) of a fixed vertex is one of the following. For $f = 1$: $(0, 0, 7)$; $f = 3$: $(0, 1, 5)$, $(1, 0, 6)$; $f = 5$: $(0, 2, 3)$, $(1, 1, 4)$; $f = 7$: $(0, 3, 1)$, $(1, 2, 2)$; $f = 9$: $(1, 3, 0)$, $(2, 0, 5)$; $f = 11$: $(2, 1, 3)$; $f = 13$: $(2, 2, 1)$; $f = 19, 21, 23$: $(3, 0, 4)$, $(3, 1, 2)$, $(3, 2, 0)$; $f = 33, 35$: $(4, 0, 3)$, $(4, 1, 1)$; $f = 51, 53$: $(5, 0, 2)$, $(5, 1, 0)$; $f = 73$: $(6, 0, 1)$; and $f = 99$ for the identity.

For $19 \le f \le 73$ the triple is the same at every fixed vertex and $a_1 + n_2 = 2f(t + r)$ equals 152, 126, 92, 198, 140, 204, 106, 146 for $f = 19$, 21, 23, 33, 35, 51, 53, 73, contradicting (4). For $f = 11, 13$ we get $\sum s_p = 22, 26$, contradicting (5). For $f = 3, 5, 7$ let k be the number of fixed vertices with $s = 1$; by (5) $3 \mid k$, and $a_1 = 30 + 2k$ for $f = 3, 5$ and $14 + 2k$ for $f = 7$, so that $3f - a_1 + 11$ is one of $-10, -16$ ($f = 3$); $-4, -10$ ($f = 5$); $18, 12, 6$ ($f = 7$), never a multiple of 7. For $f = 9$ let k be the number of fixed vertices with $s = 2$; then $\sum s_p = 9 + k$, so $k \in \{0, 3, 6, 9\}$, $a_1 = 10k$ and $3f - a_1 + 11 \in \{38, 8, -22, -52\}$, never a multiple of 7. All these cases contradict (6). Hence $f = 1$ and $(s_p, r_p, t_p) = (0, 0, 7)$. Finally, if $u \sim \sigma u$, the common neighbour of u and σu is fixed, so it is p. □

From now on σ is an involution with fixed vertex p. The 14 vertices of $N(p)$ are called *inner* and the 84 others *outer*. An outer vertex has exactly two inner neighbours, in two different blocks, and for two blocks $i < j$ the four non-adjacent pairs of inner vertices have one outer common neighbour each; so the 84 outer vertices form 42 orbits, indexed by the pairs of blocks with a sign (Figure 2). For an outer vertex u, the set $N(u) \cap N(\sigma u)$ has two elements, is σ-invariant and does not contain p: it is a single orbit $\{v, \sigma v\}$ of outer vertices.

4. The signed quotient and the frame

Choose a representative in each of the 49 orbits of size two. On the space of σ-odd functions, with basis the functions $e_o - e_{\sigma o}$, the matrix A acts by the symmetric integer matrix

$$S_{UV} = A[u, v] - A[u, \sigma v].$$

Since J annihilates odd functions, (1) gives

$$S^2 + S = 12I. \tag{7}$$

The diagonal of S is -1 on the 7 inner orbits and 0 on the 42 outer ones, so the trace is -7 and the eigenvalues are 3 and -4 with multiplicities 27 and 22. With inner representatives the first ends of the blocks and suitable outer representatives,

$$S = \begin{pmatrix} -I_7 & C \\ C^t & T \end{pmatrix},$$

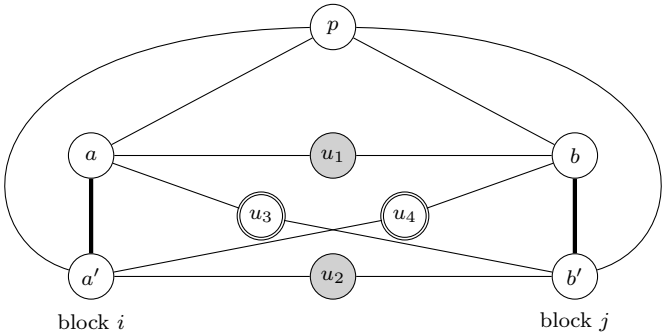


FIGURE 2. Two of the seven blocks at the fixed vertex p (thick edges) and the four outer vertices attached to them: one common neighbour for each non-adjacent pair of their ends. The involution σ exchanges a with a' and b with b', hence u_1 with u_2 (grey) and u_3 with u_4 (double circles). With the representatives a, b, u_1, u_3 these two orbits give the columns $e_i + e_j$ and $e_i - e_j$ of the matrix C of Section 4. The other adjacencies of the outer vertices are not drawn.

where the 42 columns of C are the vectors $e_i + e_j$ and $e_i - e_j$ $(i < j)$ of $\mathbb{Z}^7$, and T has zero diagonal and entries in $\{0, \pm 1\}$ with exactly ten nonzero entries in each row (an outer vertex u has 12 outer neighbours, and exactly one orbit lies in $N(u)$ entirely). Reading (7) blockwise,

$$CC^t = 12I_7, \qquad CT = 0, \qquad T^2 + T = 12I - C^tC. \tag{8}$$

Put $H = 3I - S$. Then $H^2 = 7H$ and $\operatorname{rank} H = 22$. Let V be the image of H, that is the (-4)-eigenspace of S, and $u_i = He_i/\sqrt{7} \in V$. Then $(u_i, u_j) = H_{ij}$: the seven inner vectors h_c have norm 4 and are pairwise orthogonal, the 42 outer vectors u_o have norm 3, $(h_c, u_o) = -C_{co}$, and the Gram matrix of the u_o is $3I - T$. For $v \in V$,

$$\textstyle\sum_{i=1}^{49} (v, u_i)^2 = \frac{1}{7}|Hv|^2 = 7\,(v, v), \tag{9}$$

and since $(3e_c;\, C_c^t)$ is an eigenvector of S for the eigenvalue 3 by (8),

$$3h_c + \textstyle\sum_o C_{co}\, u_o = 0 \qquad (c = 1, \dots, 7). \tag{10}$$

5. The lattice of the frame

Let N be the lattice generated by the 49 vectors u_i, let $K = \mathbb{Z}^{49} \cap V$ be the lattice of integer (-4)-eigenvectors of S, and K' the lattice of integer 3-eigenvectors. As K is primitive in $\mathbb{Z}^{49}$, its dual is the projection of $\mathbb{Z}^{49}$ on V, that is $K^* = H\mathbb{Z}^{49}/7$. Hence

$$N = \sqrt{7}\, K^*, \qquad N^* = K/\sqrt{7}, \tag{11}$$

and for $v \in N^*$ the integers $a_i = (v, u_i)$ are the coordinates of the vector $\sqrt{7}\, v$ of K, with $\sum a_i^2 = 7(v, v)$.

5.1. **The discriminant form.** For $x \in \mathbb{Z}^{49}$ we have $7x = Hx + (S + 4I)x \in K \oplus K'$, so $\mathbb{Z}^{49}/(K \oplus K') \cong \mathbb{F}_7^g$, where g is the rank of $S - 3I$ over $\mathbb{F}_7$. Thus $\det K = 7^g$, and the discriminant group of N is $K/7K^* \cong \mathbb{F}_7^{22-g}$.

Proposition 5.1. *The integer g is even, $g \le 20$, and the discriminant bilinear form of N on $\mathbb{F}_7^{22-g}$ is not split: its anisotropic kernel is a plane. Consequently N has an integral overlattice $N_{\max} \subseteq N^*$ of determinant 49 whose discriminant form is the anisotropic plane over $\mathbb{F}_7$.*

Proof. The index of $K \oplus K'$ in $\mathbb{Z}^{49}$ is odd, so over $\mathbb{Z}_2$ the lattice $\mathbb{Z}^{49}$ is the orthogonal sum of K and K', both unimodular, and the projection $c = H\mathbf{1}/7$ of the characteristic vector $\mathbf{1}$ is a characteristic vector of $K \otimes \mathbb{Z}_2$. Its norm is $(147 - s)/7$ with $s = \mathbf{1}^t S\mathbf{1}$. From (7), $588 = \mathbf{1}^t(S^2 + S)\mathbf{1} = \sum_i r_i^2 + s$, where the r_i are the row sums of S. The seven inner row sums are odd. An outer row sum is $r_o = c_o + 10 - 2m_o$, where $c_o \in \{0, 2\}$ is the column sum of C and m_o the number of entries -1 in the row of T; since $\sum_o c_o/2 = 21$ and $\sum_o m_o$ is even, the number of outer rows with $r_o \equiv 2 \pmod 4$ is odd. Hence $588 \equiv 7 + 4 + s \pmod 8$, so $s \equiv 1$ and $(c, c) \equiv 6 \pmod 8$ in $\mathbb{Z}_2$.

Write the 7-adic Jordan decomposition of K as $1^{\varepsilon_0(22-g)}\, 7^{\varepsilon g}$. The oddity formula [7, Ch. 15] (the signature plus the sum of the p-excesses is congruent to the oddity modulo 8) reads $22 + 6g + 4[\varepsilon = -1] \equiv 6 \pmod 8$. Hence g is even and $\varepsilon = (-1)^{g/2}$; as $\det K = 7^g$ exactly, $\varepsilon_0 = \varepsilon$. By (11) the discriminant form of N is the form of the unimodular constituent of K: it has dimension $2m = 22 - g$ and determinant character $\varepsilon_0 = (-1)^{g/2} = -(-1)^m$, whereas a hyperbolic form of dimension $2m$ over $\mathbb{F}_7$ has character $\left(\frac{-1}{7}\right)^m = (-1)^m$. So the form is not split; in particular $m \geq 1$, its Witt index is $m - 1$ and its anisotropic kernel is a plane. The preimage in N^* of a totally isotropic subspace of dimension $m - 1$ is the required $N_{\max}$. □

In particular the construction of Section 6 does not require the value of g.

5.2. **The dual minimum.**

Proposition 5.2. *There is no nonzero $a \in \mathbb{Z}^{49}$ with $Sa = -4a$ and $\sum a_i^2 \leq 7$. Equivalently, the minimum of N^* is at least $8/7$; equivalently, there is no nonzero integer-valued σ-odd function φ on the vertices of Γ with $A\varphi = -4\varphi$ and $\sum_x \varphi(x)^2 \leq 14$.*

Proof. If some $|a_i| \geq 2$ then $|a_i| = 2$ and the other entries have absolute sum at most 3, whereas row i of $Sa = -4a$ requires an off-diagonal sum of absolute value 8 (outer) or 6 (inner). So a has entries in $\{0, \pm 1\}$ and support of size $m \leq 7$. Lift a to an odd function φ and put $P = \{\varphi = 1\}$, $Q = \sigma P = \{\varphi = -1\}$, so $|P| = m$. Let e be the number of edges inside P and d_i the degree of $i \in P$ inside P; the equation gives that i has $d_i + 4$ neighbours in Q. Count the common neighbours of the pairs of vertices of P: the total is $m(m-1) - e$, and the vertices of $P \cup Q$ contribute $\sum_i [\binom{d_i}{2} + \binom{d_i+4}{2}] = \sum d_i^2 + 6e + 6m$. So the other vertices contribute

$$D = m(m-7) - 7e - \sum d_i^2 \ \geq 0,$$

which forces $m = 7$, $e = 0$ and $D = 0$. Then $B_{ij} = [\, i \sim \sigma j \,]$ is a symmetric 0/1 matrix on P with row sums 4, and $B^2 = 2I + 2J$ because every pair of P has its two common neighbours in Q. Its eigenvalues are 4 and $\pm\sqrt{2}$, the latter with equal multiplicities since the characteristic polynomial is rational; so the trace of B is 4, that is, four vertices of P are adjacent to their images. By Proposition 3.1 they lie in $N(p)$, and $p \notin P \cup Q$ contributes $\binom{4}{2} = 6$ to $D = 0$, a contradiction. □

6. The envelope

Theorem 6.1. *Let σ be an automorphism of order 2 of Γ. Then there are an odd unimodular lattice U of rank 24 without vectors of norm 1 and vectors $y_1, y_2, h_1, \ldots, h_7, u_1, \ldots, u_{42}$ of U such that:*

- (R1) *$(y_a, y_b) = 7\delta_{ab}$, and y_1, y_2 are orthogonal to all h_c and u_o;*
- (R2) *$(h_c, h_d) = 4\delta_{cd}$, $(u_o, u_o) = 3$, and $(u_o, u_{o'}) \in \{0, \pm 1\}$ for $o \neq o'$, nonzero for exactly ten o';*
- (R3) *$(h_c, u_o) = -C_{co}$, where the columns of C are the vectors $e_i \pm e_j$ $(i < j)$ of $\mathbb{Z}^7$;*

(R4) $3h_c + \sum_o C_{co} u_o = 0$ *for every* c;
(R5) *for every real vector* v: $\sum_c (v,h_c)^2 + \sum_o (v,u_o)^2 + (v,y_1)^2 + (v,y_2)^2 = 7\,(v,v)$;
(R6) $\mathbb{Z}y_1 + \mathbb{Z}y_2$ *is primitive in* U, *and every* $v \in U$ *outside it satisfies* $E(v) := 7(v,v) - (v,y_1)^2 - (v,y_2)^2 \geq 8$;
(R7) *for every* $v \in U$, *the* 49 *integers* (v,h_c), (v,u_o) *are the values on orbit representatives of an integer-valued* σ*-odd function* φ_v *with* $A\varphi_v = -4\varphi_v$ *and* $\sum_x \varphi_v(x)^2 = 2E(v)$;
(R8) *every nonzero vector* w *of the real span of the* h_c, u_o *whose inner products with all* h_c, u_o *are integers has* $(w,w) \geq 8/7$.

Proof. Let $N_{\max}$ be as in Proposition 5.1 and $\Pi = \mathbb{Z}r_1 + \mathbb{Z}r_2$ with $(r_a, r_b) = 7\delta_{ab}$. The discriminant form of Π is the anisotropic plane over $\mathbb{F}_7$, because -1 is not a square modulo 7. Nondegenerate forms over $\mathbb{F}_7$ are classified by dimension and determinant [19, Ch. IV, §1.7], and an anisotropic plane and its negative both have square determinant; so the discriminant forms of $N_{\max}$ and Π are anti-isometric. Gluing along the graph of an anti-isometry [18, §1], [7, Ch. 4, §3] gives an integral lattice U containing $N_{\max} \oplus \Pi$ with index 49, of rank 24 and determinant $49 \cdot 49/49^2 = 1$, in which $N_{\max}$ and Π are primitive; U is odd since it contains vectors of norm 3. Put $y_a = r_a$. Properties (R1)–(R4) are those of Section 4, (R5) is (9) together with the orthogonal basis y_1, y_2 of $V^\perp$, and (R8) is Proposition 5.2.

For $v \in U$ the projection of v on V lies in $N^*_{\max} \subseteq N^*$; by (11) its 49 inner products with the frame form a vector $a \in K$ with $\sum a_i^2 = E(v)$, and the lift of a to the vertices is the function φ_v of (R7). If $a = 0$ then $v \in U \cap V^\perp = \Pi$. Otherwise $E(v) \geq 8$ by (R8); this is (R6). In particular a vector of norm 1 would have $E \leq 7$, hence lie in Π, whose minimum is 7. □

There are exactly 156 odd unimodular lattices of rank 24 without vectors of norm 1, up to isometry [4]. We call a system of 51 vectors with the properties (R1)–(R8), for some pair (Γ, σ), a *frame* in U, and y_1, y_2 its *normals*. The pair (Γ, σ) enters only through (R7), and (R7) is used below only through Theorem 7.1. Theorem 1.1 follows from Theorem 6.1 and:

Theorem 6.2. *None of the* 156 *odd unimodular lattices of rank* 24 *without vectors of norm* 1 *contains a frame.*

The rest of this section records consequences of (R1)–(R8) that are used repeatedly.

Lemma 6.3. *In a frame, the* 42 *vectors* u_o *generate the lattice* N *generated by all* 49 *vectors* h_c, u_o. *The graph on the* u_o *in which* u_o, $u_{o'}$ *are joined when* $(u_o, u_{o'}) = \pm 1$ *is connected.*

Proof. Let M be the lattice generated by the u_o. By (R4), $3h_c \in M$. Polarising (R5) and applying it to u_o gives

$$\sum_c C_{co} h_c = -4u_o + \sum_{p \neq o} (u_o, u_p)\, u_p \in M; \tag{12}$$

the two columns of C for the pair $\{i,j\}$ give $h_i + h_j$ and $h_i - h_j$, hence $2h_i \in M$ and $h_i \in M$. Now define $T_{oo'} = -(u_o, u_{o'})$ for $o \neq o'$ and $T_{oo} = 0$. Pairing (12) with $u_{o'}$, $o' \neq o$, gives $\sum_{o''} T_{oo''} T_{o''o'} + T_{oo'} = -(C^t C)_{oo'}$. If the graph were the disjoint union of two nonempty parts, the left side would vanish for o, o' in different parts, whereas $(C^t C)_{oo'} = \pm 1$ when o and o' share exactly one block. So two orbits sharing exactly one block lie in the same part; but the graph "sharing exactly one block" on the 42 orbits is connected. □

Lemma 6.4. *A lattice containing a frame is indecomposable.*

Proof. Let $U = L_1 \oplus L_2$ with $L_1, L_2 \neq 0$. The summands are unimodular without vectors of norm 1, so a vector of norm 3 lies in one of them, and vectors of different summands are orthogonal. By Lemma 6.3 all the u_o lie in one summand, say L_1, and so do the h_c. Then L_2

is orthogonal to the 49 vectors, hence contained in the plane of y_1, y_2; but a unimodular lattice of rank 1 or 2 contains a vector of norm 1. □

Definition 6.5. Let U be an odd unimodular lattice of rank 24 and U_0 its sublattice of vectors of even norm. The quotient U_0^*/U_0 has four classes: U_0, $U \setminus U_0$ and two classes S_1, S_2, whose union is the *shadow* $\{z/2 : z \text{ characteristic}\}$ of U. The vectors of norm 2 of the shadow are the *new roots* of U: they are the roots of the even unimodular lattices $U_0 \cup S_j$ that are not in U. (For the shadow and its properties see [8].)

Lemma 6.6. *Let β be a new root of a lattice containing a frame. Then $w(\beta) := \sum_c (\beta, h_c)^2 \in \{1, 3\}$, the integers (β, h_c) have absolute value at most 1, and at most one of the 44 vectors u_o, y_a has $|(\beta, \cdot)| = 3/2$; if there is one then $w(\beta) = 1$, otherwise $w(\beta) = 3$.*

Proof. With $z = 2\beta$ characteristic, the products (z, u_o) and (z, y_a) are odd and (z, h_c) is even. In (R5) for $v = \beta$ the right side is 14; the 44 terms from u_o, y_a are at least $1/4$ each, a product of absolute value $3/2$ adds 2 and a larger one at least 6. Hence $w(\beta) = 3 - 2 \cdot \#\{\text{products equal to } \pm 3/2\} \geq 0$. □

Lemma 6.7. *For a characteristic vector z of norm 16 of a lattice containing a frame, $(z, y_1)^2 + (z, y_2)^2 \leq 66$.*

Proof. $E(z) = 112 - Y$ with $Y = (z, y_1)^2 + (z, y_2)^2 \equiv 2 \pmod 8$, and $E(z) = \sum_o (z, u_o)^2 + 4\sum_c b_c^2$ with $(z, h_c) = 2b_c$ and the (z, u_o) odd. Modulo 8 this gives $6 \equiv 42 + 4\sum b_c^2$, so $\sum b_c^2$ is odd, $E(z) \geq 46$ and $Y \leq 66$. □

Lemma 6.8 (Twins)**.** *Let y_1, y_2 be the normals of a frame in U, $M = U \cap \{y_1, y_2\}^\perp$ and $\Pi = \mathbb{Z}y_1 + \mathbb{Z}y_2$. Let U' be any unimodular lattice containing $M \oplus \Pi$ in which M and Π are primitive. Then U' is odd, has no vector of norm 1, and the same 51 vectors form a frame in U'.*

Proof. The frame vectors lie in $M \oplus \Pi$, so (R1)–(R5) and (R8) hold in U'. As M is primitive in the unimodular lattices U and U', both project onto M^* in the space V spanned by M. Since $E(v)$ is 7 times the norm of the projection of v on V, and the function φ_v depends only on that projection, (R6) and (R7) pass from U to U'. A vector of norm 1 of U' would have $E \leq 7$ and lie in Π. □

7. Energy and eigenfunctions of small norm

By (R5) the energy $E(v)$ of $v \in U$ is 7 times the squared norm of the projection of v on the space of the frame. It is a non-negative integer, it is unchanged when a multiple of y_1 or y_2 is added to v, and it vanishes exactly on $\mathbb{Z}y_1 + \mathbb{Z}y_2$. By (R7), for $E \geq 1$ a statement "there is no integer σ-odd eigenfunction of squared norm $2E$" forbids every vector of energy E in every lattice containing a frame.

Theorem 7.1 (Lemmas T_{16}, T_{18}, T_{20})**.** *Let σ be an automorphism of order 2 of Γ. There is no integer-valued function φ on the vertices of Γ with $\varphi \circ \sigma = -\varphi$, $A\varphi = -4\varphi$ and $\sum_x \varphi(x)^2 \in \{16, 18, 20\}$.*

The three proofs are long and computer-assisted; they are given in full in the supplement [21] (`lemma/T16_PROOF.md`, `T18_PROOF.md`, `T20_PROOF.md`). They follow the scheme of Proposition 5.2. First, φ takes the values $0, \pm 1$. With $S^+ = \{\varphi = 1\}$ and $S^- = \sigma S^+$, each vertex of S^+ has four more neighbours in S^- than in S^+, and the count of common neighbours bounds the number of edges inside S^+. The bipartite graph between S^+ and S^- is then one of finitely many *support structures*: for the squared norm 16, the complements of the cube and of the

Wagner graph; for 20, nine configurations, sorted by the number of support vertices adjacent to p. Each structure is excluded, by hand where possible, and otherwise by an exhaustive search over the vertices outside the support in which every rejection is exact: a count of common neighbours; an integer vector on which one of the positive semidefinite matrices $9(3I - A) + J$ and $11(4I + A) - 2J$ would take a negative value; or, for one family of T_{18} and at sixteen nodes of the first program of T_{20}, an exact computation of a rank that would exceed 22 (the σ-odd part of $9(3I - A) + J$ has rank at most 22). For T_{20} the second program closes these sixteen nodes by its own search, without the rank bound. The largest group of searches of T_{20} visits $1.58 \cdot 10^7$ nodes from 615,405 starting configurations. How these computations were verified is described in Section 9.

Corollary 7.2. *Let U contain a frame. Then $E(v) \geq 11$ for every $v \in U$ outside $\mathbb{Z}y_1 + \mathbb{Z}y_2$.*

Adding multiples of y_1, y_2 we may assume $|(v, y_a)| \leq 3$; then $(v, v) = (E(v) + a^2 + b^2)/7$ with $a = (v, y_1)$, $b = (v, y_2)$ and $a^2 + b^2 \leq 18$, so $(v, v) \leq 4$ when $E(v) \leq 10$. Going through the norms 2, 3, 4 shows that the vectors of energy 8, 9, 10 are, up to $\mathbb{Z}y_1 + \mathbb{Z}y_2$, exactly the following *witnesses*:

(W8) a vector of norm 3 with $\{|a|, |b|\} = \{2, 3\}$;
(W9) a root with $\{|a|, |b|\} = \{1, 2\}$;
(W10) a root with $\{|a|, |b|\} = \{0, 2\}$, or a vector of norm 4 with $|a| = |b| = 3$.

In the same way a vector of energy at most 7 outside $\mathbb{Z}y_1 + \mathbb{Z}y_2$ has a representative of norm 2 or 3. So (R6) and Corollary 7.2 are finite tests on the vectors of norm 2, 3, 4 of U: a pair (y_1, y_2) with a witness is the pair of normals of no frame, and exhibiting the witness proves it. The numbers of vectors of these norms satisfy $N_4 = 196560 - 24N_2 - 24N_3$, and $N_3 = 4096$ [8]; the first identity is used to certify that the lists are complete.

Corollary 7.3. *Let U contain a frame. Then every root r of U has $|(r, y_1)| \leq 1$ and $|(r, y_2)| \leq 1$. In particular, if U has a root r with $(r, y) = \pm 2$ for some vector y of norm 7, then y is a normal of no frame of U, whatever the other normal.*

Proof. $E(r) = 14 - a^2 - b^2 \geq 8$ gives $a^2 + b^2 \leq 6$; if $|a| = 2$ then $E(r) = 10 - b^2 \in \{9, 10\}$, contradicting Corollary 7.2. □

8. The 156 lattices

This section describes the proof of Theorem 6.2. Lattices are referred to by their number 000–155 in the table of Appendix A, in which they are ordered by increasing number of roots, and by their root system.

8.1. **The lattices.** The 156 lattices are given by generators. For each of them the following was computed twice, first by programs with floating-point steps (in the enumeration of short vectors and in the ranks of the root components), then by an independent program in exact arithmetic: the determinant is 1, the lattice is odd, it has no vector of norm 1, and its root system is the one of the classification. The root system separates all but seven pairs of lattices, and those are separated by their two even neighbours (also recomputed exactly). So the data are 156 pairwise non-isometric lattices of the list, hence all of them by [4].

8.2. **Twenty lattices excluded without enumeration.**

Proposition 8.1 (Packing)**.** *Let U contain a frame, let $j \in \{1, 2\}$ and let R_j be the set of new roots in the class S_j. For a vector x of norm 3 let $m_j(x)$ be the number of $\beta \in R_j$ with $|(\beta, x)| = 3/2$. Then:*

(a) $\sum_{o=1}^{42} m_j(u_o) \le |R_j|$. *In particular the sum of the* 42 *smallest values of* m_j *on the vectors of norm* 3 *taken up to sign is at most* $|R_j|$*, and a vector* x *of norm* 3 *can be one of the* $\pm u_o$ *only if* $m_j(x)$ *plus the sum of the* 41 *smallest values of* m_j *on the other classes is at most* $|R_j|$ (x *is then called* admissible).

(b) *If* $\mathcal{A}$ *is a set of vectors of norm* 3 *containing all the* u_o *and* L *the lattice it generates, then* $\operatorname{rank} L \ge 22$*, and the vectors* w *of the real span of* L *with* $(w, L) \subseteq \mathbb{Z}$ *and* $(w, w) < 8/7$ *span, together with* $L^\perp$*, a space of dimension at most* 2.

Proof. (a) By Lemma 6.6 a new root is met at $\pm 3/2$ by at most one u_o, so the 42 sets counted by the $m_j(u_o)$ are pairwise disjoint subsets of R_j; and the u_o are pairwise distinct up to sign by (R2). (b) By Lemma 6.3, L contains N, of rank 22. Let w be such a vector. It has integer products with the h_c and u_o, and so has its projection on the span V of the frame, whose norm is less than 8/7; by (R8) this projection is zero, so $w \in V^\perp$. Also $L^\perp \subseteq N^\perp = V^\perp$, which has dimension 2. □

For every lattice one has $m_1 = m_2$ on the vectors of norm 3 (if $(\beta, x) = 3/2$ then $x - \beta$ is a new root of the other class with the same product). Table 1 gives the outcome of the computation for 18 lattices: for fourteen of them the first inequality of (a) fails; for the four others the lattice generated by the admissible vectors violates (b).

lattice	roots	Σ_{42}	$\min \lvert R_j \rvert$	lattice	roots	Σ_{42}	$\min \lvert R_j \rvert$
094	d_5^4	84	80	147	$a_{15}d_8$	84	32
128	$a_9e_6d_6a_1$	84	64	148	$a_{15}e_7a_1$	168	64
129	$e_6^2d_5^2$	84	64	150	$d_8e_7^2a_1^2$	168	64
133	$a_{13}d_6a_3a_1$	168	128	151	$d_{12}d_8d_4$	336	128
136	a_{11}^2	84	48	152	$e_8d_8^2$	672	256
142	$d_{10}d_6^2a_1^2$	168	128	154	a_{23}	252	48
144	$a_{16}a_7$	168	104	155	$d_{16}d_8$	672	128

lattice	roots	dimension	lattice	roots	dimension
117	$d_8d_4^4$	8	145	$d_8^2d_8$	8
134	$d_9a_7^2$	9	146	d_8^3	24

TABLE 1. The packing test. Above: the sum Σ_{42} of the 42 smallest values of m_j exceeds the number of new roots of a class. Below: the dimension spanned by the short dual vectors of Proposition 8.1(b), for $\mathcal{A}$ the set of admissible vectors, exceeds 2.

Proposition 8.2. *The lattices* 153 (*root system* d_{12}^2) *and* 152 (*root system* $e_8d_8^2$) *contain no frame.*

Proof. The lattices $D_{12}^+ \oplus D_{12}^+$ and $E_8 \oplus O_{16}$, where D_{12}^+ and O_{16} are the odd unimodular lattices of rank 12 and 16 with root systems d_{12} and d_8^2, are odd unimodular of rank 24 without vectors of norm 1, with root systems d_{12}^2 and $e_8d_8^2$. Each of these root systems occurs once in the list, so these are the lattices 153 and 152. They are decomposable; apply Lemma 6.4. □

Proposition 8.3. *The lattice* 143 (*root system* $a_{15}a_8$) *contains no system of vectors satisfying* (R2) *and* (R5).

Proof. Summing (R5) over an orthonormal basis of a subspace W gives $\sum_f |P_W f|^2 = 7 \dim W$, the sum being over the 51 frame vectors and P_W the orthogonal projection. Let W be the

span of the component a_8 ($\dim W = 8$) and ℓ the line orthogonal to all roots. The vectors of U orthogonal to the roots are the multiples of a primitive vector z with $(z,z) = 144$, and the 4096 vectors x of norm 3 are of three kinds:

$(x,z)^2$	$\lvert P_W x\rvert^2$	$\lvert P_\ell x\rvert^2$	number
81	0	9/16	1120
169	8/9	169/144	288
9	2	1/16	2688

In each case $4|P_W x|^2+16|P_\ell x|^2 \geq 9$. Summing over the 42 vectors u_o gives $378 \leq 4\cdot 56+16\cdot 7 = 336$, a contradiction.

The table follows from the description of U as a gluing of $A_{15} \oplus A_8 \oplus \mathbb{Z}z$: writing $x = x_{15} + x_8 + \frac{t}{144}z$ with $t = (x,z)$, the component x_{15} lies in the class $5t$ of $A_{15}^*/A_{15} \cong \mathbb{Z}/16$ and x_8 in the class $7t$ of $A_8^*/A_8 \cong \mathbb{Z}/9$ (for suitable generators), the norms in the class i of A_n^* are $i(n+1-i)/(n+1)+2k$, $k \geq 0$, and among the $|t| \leq 20$ norm 3 occurs exactly for $t = \pm 3$ ($\frac{15}{16}+2+\frac{1}{16}$), $t = \pm 9$ ($\frac{39}{16}+0+\frac{9}{16}$) and $t = \pm 13$ ($\frac{15}{16}+\frac{8}{9}+\frac{169}{144}$), with $2\cdot 16\cdot 84$, $2\cdot 560$ and $2\cdot 16\cdot 9$ vectors. The details, and the exact verification of the gluing on the data, are in `lemma/upstream/LATTICE_143.md` of [21]. □

Propositions 8.1, 8.2 and 8.3 exclude the 20 lattices 094, 117, 128, 129, 133, 134, 136, 142–148, 150–155; lattice 152 is covered twice. (In Appendix A the codes EP, SP, So recall that some of these lattices were first excluded by other arguments, which are not used here.)

8.3. **Enumeration of the pairs of normals.** For the other 136 lattices the pairs (y_1, y_2) of orthogonal vectors of norm 7 are enumerated up to automorphisms of U and signs. Only pairs satisfying the following necessary conditions are produced; they are instances of (R6) and of Lemma 6.6: for every root α, $(\alpha,y_1)^2+(\alpha,y_2)^2 \in \{0,1,2,4,5\}$; for every vector x of norm 3, $|(x,y_a)| \leq 3$ and not both equal to 3; for every new root β, $|(\beta,y_a)| \leq 3/2$ and not both equal to 3/2; and no nonzero vector of the dual of $U \cap \{y_1,y_2\}^\perp$ has norm less than 8/7.

Reflections in roots are automorphisms of U. So y_1 may be taken dominant for the Weyl group $W(R)$ of the root system, and then y_2 dominant for the stabiliser of y_1 in $W(R)$, which is generated by the simple reflections fixing y_1 [11, §1.12]. The dominant vectors are obtained from their projections on the root components (dominant weights with labels at most 2, by the condition on roots, combined according to the glue classes) and from their component in the orthogonal complement of the roots, enumerated by a Fincke–Pohst procedure [10] with exact final tests. The pairs are further reduced by a group of diagram automorphisms, each verified to preserve U; using a subgroup of the automorphism group is always sound. We call this procedure the *main enumeration*; it is used for 102 lattices. The conditions for a frame are symmetric in y_1, y_2 and the enumeration contains both orders, so a pair is skipped when $I(y_1) > I(y_2)$, where I is an invariant of the orbit of a vector (the sum of $(\alpha,y)^2$ over the roots, and the sorted numbers of new roots of the two classes met at 3/2). For the odd Leech lattice (000) and the lattices 001, 002, 004, 010, with root systems a_1^8, a_1^{12}, a_1^{16}, a_1^{24}, the pairs are produced by separate programs: 141 classes for the odd Leech lattice, and lists in coordinates of the Niemeier lattice with root system A_1^{24} for the four others.

The completeness of the pair enumeration was compared with an independent enumerator (all vectors of norm 7 by Fincke–Pohst, its own Weyl reduction, no diagram automorphisms) on four lattices: 140, 135, 141 and 064, the last one with roots that do not span the space. The two sets of pairs coincide in each case (288, 464, 331 and 1,050,528 pairs). For this comparison the producer was run without diagram automorphisms; the reduction by these automorphisms, used in the main enumeration, is not covered by it: it rests on the verification

of each automorphism (it preserves the Gram matrix and permutes the simple roots) and on the few lines of the program that select one representative per orbit. For the other lattices completeness rests on the argument above and on the program that produces the pairs.

8.4. **Exclusion of a pair.** Each pair is excluded in one of the following ways.

(i) At once. A vector of energy at most 7, contradicting (R6); a witness (W8), contradicting T_{16}; or a characteristic vector of norm 16 violating Lemma 6.7.

(ii) By a witness (W9) or (W10), contradicting T_{18} or T_{20}. Each such pair has a certificate: the vectors y_1, y_2 and the witness in coordinates; a separate program checks in exact arithmetic that the three vectors lie in U and have the required norms and inner products.

(iii) By search. The candidates for the h_c are the vectors of norm 4 orthogonal to y_1, y_2 with $|(\beta, h)| \leq 1$ for every new root β. By Lemma 6.6 every new root β must be *covered*: $w(\beta) = 1$ if a normal meets β at 3/2, and $w(\beta) \in \{1, 3\}$ otherwise, with $w(\beta) = 1$ only if some u_o meets β at 3/2. The search looks for seven pairwise orthogonal candidates with these coverages, branching on a new root that still needs covering, and discards a partial choice as soon as a necessary condition fails. The conditions used are consequences of (R1)–(R5) and of the lemmas of Section 6. The main ones are: for every pair of chosen h there must remain candidates for both vectors u_o of that pair of blocks (vectors of norm 3, orthogonal to y_1, y_2 and to the other chosen h, with the products (R3)); $h_1 + \cdots + h_7 \in 2U$ (sum (R4) over c: every column of C has even sum); for each c the twelve vectors u_o with $C_{co} \neq 0$, suitably signed, sum to $3h_c$, which constrains their products with every new root; and finally the 42 vectors must be distinct, with a Gram matrix $3I - T$ satisfying (8). Reflections of U fixing the vectors already chosen are used to identify equivalent branches. A search that exceeds a preset number of nodes is reported as incomplete and is not an exclusion; no pair was left in that state.

(iv) By transfer to a twin. Let U' be the unimodular lattice of Lemma 6.8 obtained by composing the gluing map of U with the isometry $\left(\begin{smallmatrix} 2 & 2 \\ -2 & 2 \end{smallmatrix}\right)$ of the discriminant form of $\mathbb{Z}y_1 + \mathbb{Z}y_2$; its roots are obtained from the vectors of norm 2, 3, 4 of U. The pair (y_1, y_2) hosts a frame in U if and only if it does in U', and U' is a lattice of the list. The program uses this in three forms, tried in this order.

(a) *Status.* If all the lattices of the list with the root system of U' were already recorded as containing no frame when the pair was treated, the pair is excluded.
(b) *More roots.* Otherwise, if U' has strictly more roots than U, the pair is set aside, and it is excluded when all the lattices with more roots than U are. This is a descending induction on the number of roots; it does not require U' to be identified, only its number of roots, as computed by the program, to exceed that of U. The dependencies it creates between lattices (1,622 of them) contain no cycle.
(c) *Same lattice.* Otherwise, if U' has the root system of U and this root system occurs once in the list, then $U' \cong U$, and the pair is replaced by its image in U when an invariant key of the pair decreases strictly.

Forms (a) and (c) need the root system of U'; the program reads it from the numbers of roots of the components and from their ranks, computed in floating point. Form (a) moreover depends on the order in which the lattices were treated. What this implies for the proof is discussed in Section 9.

The correspondence between these rules and the program, rule by rule, with the derivation of each rule from the properties of a frame, is the audit `lemma/upstream/AUDIT_ENGINE_RULES.md` of [21].

8.5. **Results.** Table 2 summarises how the 156 lattices are treated; the paragraphs below give the details, and Appendix A the numbers lattice by lattice.

treatment	lattices	pairs
arguments on the lattice (Propositions 8.1, 8.2, 8.3)	20	—
main enumeration; exclusion at once, by search or by transfer	91	220,260,576
the same, and witnesses (W9), (W10) with certificates	13	373,439,131
odd Leech lattice: linear certificates and searches	1	141 classes
lattices 004, 010: lists of pairs	2	420,345
lists of pairs (September), then lemmas and a complete second run	29	6,713,853
	156	

TABLE 2. The treatment of the 156 lattices.

91 *lattices of the main enumeration.* For these lattices every pair goes through (i), (iii) or (iv); the enumeration itself does not use the lemmas T_{18}, T_{20}. They enter for these lattices in one way only: for the representatives y_1 whose records come from a program version that was not recorded and that were not run again, the exclusion rests on Corollary 7.3 (Section 9, item 3 of the paragraph on the search). The records were recounted: every representative y_1 is covered exactly once, and there are 220,260,576 pairs, of which 97,628,147 are excluded at once, 22,702,398 by search, 6,107,330 by transfer, 93,822,540 are skipped by the symmetry in (y_1, y_2), and 161 were first reported incomplete and then excluded by a second search with a larger limit.

The 13 *lattices with the smallest root systems.* (001–003, 005–009, 011–015). Here the searches are long, and a pair with a witness (W9) or (W10) is set aside instead of searched: 5,190,752 pairs are excluded through T_{18} and 2,508,295 through T_{20}, each with its certificate (ii); the other pairs are excluded by (i), (iii), (iv). For one representative y_1 of lattice 003 ($a_2^2 a_1^{10}$) the enumeration of the y_2 did not terminate; this y_1 has a root r with $(r, y_1) = -2$ and is excluded by Corollary 7.3.

The odd Leech lattice and the lattices 004*,* 010*.* The 141 classes of pairs of the odd Leech lattice are excluded by 86 certificates of infeasibility of a linear program, verified in integer arithmetic, and 55 searches. The pairs of 004 and 010, 416,383 and 3,962 in number, are excluded by (i) and (iii).

The remaining 29 *lattices.* These were first treated in September 2026 on lists of pairs by earlier versions of the search. They were all treated again on 7 October 2026, in two ways.

By the lemmas alone, for 28 *of them.* Every admissible y_1 up to the Weyl group was listed (4,334 in all). Of these, 4,073 have a root with product ± 2 and are excluded by Corollary 7.3. For the 261 others all pairs were listed: 173,708 pairs, of which 171,056 have a vector of energy at most 10, stored with the pair and verified by an exact checker. This leaves 2,652 pairs without witness, in 11 lattices. The 17 lattices 087, 096, 108, 115, 116, 119, 121, 122, 125, 127, 132, 135, 137–141 are thus excluded by Theorem 7.1 and explicit witnesses, without any search.

By a complete second run. The 28 lattices were run again in full with the final version of the program: 6,657,309 pairs, all excluded. For 27 lattices (1,168,349 pairs, of which 85,080

are closed by the transfer rule) each job was run on two machines, with identical records. For lattice 044 (a_3^8), the largest (5,488,960 pairs, of which 345,792 are closed by the transfer rule), each job was run once. The transfer rule is not needed for these 28 lattices: every pair is covered by Corollary 7.3, or has a witness of energy at most 10, or is one of the 2,652 pairs without witness; and of these, 1,610 are excluded by Lemma 6.7, 1,004 by search, and 38, which the run had closed by transfer, by a separate run with the transfer rule switched off. The 29th lattice, 105 ($e_6d_5a_5^2a_1^2$), was run again in the same way: 56,544 pairs, all excluded.

9. Verification, reproducibility, and what the proof rests on

We follow the principle that a computed step is trusted when an independent program reproduces it, or when it produces a certificate that a simple separate program checks. This section states for each part which of the two holds, and where neither does.

The mathematical reduction. Sections 3–6 are proved in full in this paper. They use two cited theorems: the oddity formula and the classification of quadratic forms over $\mathbb{F}_7$.

The three lemmas. Each of T_{16}, T_{18}, T_{20} consists of reductions by hand and of finite exhaustive searches. For T_{16} the searches produce explicit obstructions (240, and 32 witnesses for a second case), which are checked by a separate verifier and were recomputed by an independent program. For T_{18} every computation was done by at least three programs, one of them written from the text alone. For T_{20} every computed step has a second program; for one of the nine configurations the second program shares the description of the model and part of the enumeration skeleton with the first, and in the large searches the second program traverses the tree of the first again, with the same choice of the object to branch on and its own tests. Each of the three texts was refereed separately from its writing.

The lists of short vectors. For all 156 lattices the lists of vectors of norm 2, 3, 4 built by the final version of the program were verified in exact arithmetic, on one machine: every vector has its norm, the vectors are distinct, and the counts satisfy the identity $N_4 = 196560 - 24N_2 - 24N_3$. Since a list can only be too short, these three lists are complete. The numbers of vectors of norm 4 and of new roots recorded by the machines of 7 October agree with them on the 22 lattices where they were compared. The lists built by the earlier versions of the program were not re-examined.

The transfer rule. Over the 102 lattices of the main enumeration (592,864,217 pairs) the three forms of Section 8.4(iv) close 1,300,094 pairs (status), 32,265,417 pairs (more roots) and 2,863,005 pairs (same lattice). The second form is justified by induction on the number of roots; it uses the number of roots of the twin as the program counts it, by integer tests on vectors whose coordinates are formed in floating point; an exact recount has not been made. The two others, 4,163,099 pairs in all, rest on the recognition of the root system of the twin lattice through ranks computed in floating point, and the first of them on the order in which the lattices were treated during the campaign, as recorded at the time. The arithmetic of the recognition was compared with an integer computation on all the pairs of two lattices (112 and 114; 40,652 pairs): the two agree on every pair; these two lattices have no pair closed by transfer, so this does not test the dependencies. Corollary 7.3 removes a part of the question: 42,276 of the 54,096 representatives y_1 of the 102 lattices have a root with product ± 2 and are excluded whatever the verdicts of their pairs, which leaves at most 2,479,669 pairs depending on the first or the third form. A recertification of these pairs that uses neither a recorded status nor floating-point arithmetic (exact roots of the twin; then more roots, or strict descent in the same lattice, or a search without the transfer rule) has not been carried out.

The enumeration of the pairs. One program, compared with an independent enumerator on four lattices (Section 8.3); for the others it rests on the program that produces the pairs, and the reduction by diagram automorphisms is not covered by the comparison. After the fact that no human expert has read the proof, this is its main reservation. The runs of 7 October on 29 lattices (last paragraph of Section 8.5) use no diagram automorphism.

Certificates. The 7,762 records of pairs excluded through T_{18} and T_{20} in the main enumeration all have certificates recomputed from regenerated lists of pairs, and the 17,983 such pairs of the lattices 001, 002 were checked separately. The witnesses of the 171,056 pairs of the 28 lattices and the 4,073 certificates for Corollary 7.3 were verified by two checkers written independently.

The search. The exclusions by search, about $2.6 \cdot 10^7$ pairs in all, are the output of one program and carry no certificate. What supports them:

(1) Every rule of the program was derived from the properties of a frame in a written audit, by a party that had not written the program.
(2) The audit found five conditions under which a verdict could be wrong although the rules are sound. Each was discharged: one by a census of all records (the condition never occurred); one by an exact re-examination of 147,475,940 pairs of the 25 lattices concerned (the 8,432 pairs exposed to it were all excluded again by the complete version of the search; for three representatives y_1 of lattice 003 this re-examination is replaced by Corollary 7.3); one, the completeness of the lists of short vectors, by the verification described above, within the limits stated there; one by running a lattice again without the test concerned; and one by Theorem 7.1 for the squared norm 16.
(3) Records of September whose program version had not been archived or recorded concerned 10,802 representatives y_1 in 55 lattices of the main enumeration. Of these, 9,143 are excluded independently by Corollary 7.3, with certificates; the program used for 6,282 of them was recovered and archived; and the 713 representatives that had neither were run again with the final version (for sixteen lattices the whole lattice was run again): 2,419,830 pairs, each job on two machines, identical records; 60,308 of these pairs are closed by the transfer rule, so that for them the new record still points to other lattices. The 28 lattices of the last paragraph of Section 8.5 were run again in the same conditions, lattice 044 on one machine only.
(4) The records of the lattices 000, 004 and 010 (27 September to 2 October 2026; those of 004, 010 and of 47 classes of 000 with archived versions of the program) were not produced again; those of 004 and 010 were recounted.

A second, independently written search program does not exist. For two lattices (115, 132) and for 86 classes of the odd Leech lattice the records of September also hold exclusions by programs of two other parties (a linear program with integer certificates; a separate check of the seven vectors h_c); these were not re-examined for this paper.

The packing test. Table 1 was computed by two programs written independently, the second one in exact arithmetic from the generators of the lattices; the upper part of the table was computed a third time. For the four lattices of the lower part, all 2,048 classes of vectors of norm 3 are admissible, they form a single connected component, of rank 24, and the vectors of Proposition 8.1(b) (8, 9, 8 and 24 of them up to sign, all of norm 1) are linearly independent.

Computing environment. The final version of the programs runs under Python 3.12 with numpy 2.5, numba 0.67 (compiled inner loops of the search and of the enumerations) and python-flint 0.9 (exact linear algebra). The main enumeration ran on rented machines with

x86-64 processors of two makes (4 to 128 cores) and, in September, on one desktop computer. The job records of the 102 lattices of the main enumeration total about 6,700 hours of elapsed time, each job using one or two cores, except 8,001 jobs with several worker processes, which total 1,058 worker-hours. These figures do not include the interrupted jobs, the lattices 000, 001, 002, 004, 010, the runs of September on lists of pairs, or the verification runs; a total in core-hours is not available.

Possible sources of error. As in [15], the result depends on long computations whose execution cannot be followed by a reader, and it is right to say where an error could hide. A *mathematical* error in the rules would show in the written derivations of Sections 6–8 and in the audit of the program against them. A *programming* error is what the second programs, the certificates and the comparisons of this section are for; they cover the lemmas, the short vectors, the lattices themselves, the pairs excluded by a witness and, on four lattices, the enumeration; they do not cover the searches, which rest on one program. An undetected *hardware* error cannot affect a certificate or a witness, which are checked afterwards on another machine, nor the jobs that were run twice with identical records; most jobs of the main enumeration were run once.

Summary of what the theorem rests on.

(1) Cited theorems: the oddity formula [7]; the classification of the odd unimodular lattices of rank 24 [4]; the identity between N_2, N_3, N_4 (theta series of unimodular lattices, [7]).
(2) The three lemmas of Theorem 7.1, computer-assisted, verified as described.
(3) The completeness of the enumeration of the pairs, compared with an independent enumerator on four lattices.
(4) The correctness of the search program: audited and run again, not duplicated.

The whole proof, including the hand arguments of this paper, was produced with the assistance of AI systems (see the Acknowledgements), working in separate sessions and checking one another's work. At the time of writing it has not been read by an independent expert, and no part of it has been formalised in a proof assistant; a formalisation of Sections 3–6 and of the certificate checkers is planned.

Reproducibility. The repository [21] contains the programs, the commands, the records of every run, the certificates and their checkers, the referee reports, and a table giving for each computation cited here the program version, the location of its output and the line to read.

10. Concluding remarks

The automorphism group. With the exclusions of the orders 3, 7 and 11 quoted in the Introduction, Theorem 1.1 leaves no possible symmetry: a strongly regular graph with parameters $(99, 14, 1, 2)$ can only be asymmetric (Corollary 1.2 and [14]). An existence or nonexistence proof can therefore no longer start from a prescribed automorphism.

A proof without computer. Theorem 6.1 is elementary, and so are the exclusions of twenty lattices in Section 8. We do not know a proof of Theorem 6.2 without computer. The three lemmas of Theorem 7.1 say that an integer σ-odd eigenfunction for the eigenvalue -4 has squared norm at least 22; by themselves they exclude seventeen more lattices, and every further step of this kind would shorten the searches.

Formalisation. The exclusions of the orders 3 and 7 are formalised [12, 13]. For order 2 a natural first step is Sections 3–6, with Theorem 7.1 and Theorem 6.2 as named hypotheses, and then the checkers of the certificates.

The open computational item. The recertification of the pairs closed by the first and the third form of the transfer rule (Section 9) remains to be done.

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This work was carried out with the assistance of AI systems (large language models), which were used to develop the arguments, to write and check the programs, and to draft and cross-check the texts. The author directed the work and takes full responsibility for its content.

Appendix A. The 156 lattices

In the following table the number of pairs is the number of pairs produced by the enumeration of Section 8.3, both orders included. For the rows marked † it is the number of pairs of the run of 7 October, in which no diagram automorphism is used; these numbers are not comparable with the others and are not in the totals. Rows without numbers are lattices for which no pair is enumerated, or for which no recounted report gives the number.

Table 3: The 156 lattices. Method: S stream sweep; SD stream sweep with pairs through T_{18}, T_{20}; GD, G Golay pair files; O the odd Leech lattice; H, F, H3 pair lists of 27–29 September; P, EP, SP packing (Proposition 8.1; EP and SP: also by earlier arguments not used here); So decomposable lattice (Proposition 8.2); B budget identity (Proposition 8.3). †: pairs of the whole-lattice re-run with the archived engine (identity symmetry).

id	roots	method	pairs	by search	via T_{18}	via T_{20}
000	—	O				
001	a_1^8	GD	117,068		4,545	3,169
002	a_1^{12}	GD	718,422		6,675	3,594
003	$a_2^2a_1^{10}$	SD	17,757,312	21,864	39,642	43,210
004	a_1^{16}	G				
005	$a_2^4a_1^8$	SD	54,636,369	59,248	297,680	253,904
006	$a_3^2a_1^{12}$	SD	7,051,414	213,134	72,717	30,385
007	$a_3a_2^4a_1^6$	SD	62,629,395	86,849	817,065	482,534
008	a_2^8	SD	10,026,322	20,538	152,429	93,865
009	$a_2^6a_1^6$	SD	18,780,847	696,879	101,399	53,011
010	a_1^{24}	G				
011	$a_3^2a_2^4a_1^4$	SD	108,539,672	476,117	1,971,590	909,188
012	$a_2^8a_1^4$	SD	2,624,309	185,244	0	0
013	$a_3^4a_2^2a_1^2$	SD	21,063,451	910,033	64,761	9,952
014	$a_4a_3a_2^4a_1^4$	SD	37,020,901	180,047	984,326	334,300
015	$a_3^2a_3a_2^4a_1^2$	SD	32,473,649	223,276	677,923	291,183
016	$a_3^4a_1^4a_1^4$	S	9,661,027	353,773		
017	$a_3^4a_1^8$	S	1,193,626	104,715		
018	$a_4a_3^2a_3a_2^2a_1^2$	S	57,409,330	5,355,985		
019	$a_4^2a_2^4a_1^4$	S	12,492,920	1,210,159		
020	a_3^6	S	1,080,309	106,281		
021	$a_3^4a_2^4$	S	6,720,913	638,194		
022	$d_4a_4a_2^6$	S	3,719,056	411,475		
023	$d_4a_3^4a_1^4$	S	2,902,784	344,783		
024	$a_5a_3^3a_1^6a_1$	S	3,331,604	387,198		
025	$a_4^2a_3^2a_3a_1^2$	S	12,685,851	1,316,548		

id	roots	method	pairs	by search	via T_{18}	via T_{20}
026	$a_5a_3^2a_2^4a_1$	S	5,525,802	597,257		
027	$d_4a_3^4a_1^4$	S	1,858,950	202,485		
028	$d_4^2a_1^{16}$	S	50,646	8,088		
029	$a_4^2a_3^2a_2^2a_1^2$	S	20,342,381	2,103,221		
030	$a_3^4a_3^2a_1^4$	S	1,460,380	166,827		
031	$a_4^4a_1^4$	S	2,051,633	221,699		
032	$a_5a_4a_3a_3a_2a_2a_1$	S	19,608,530	2,108,197		
033	$a_4^2a_3^4$	S	2,049,478	216,006		
034	$a_4^2a_4a_3a_2^2a_1^2$	S	7,280,902	793,546		
035	$a_5a_4^3a_1^3$	S	1,658,601	198,640		
036	$a_5^2a_3^2a_2^2$	S	2,379,274	270,033		
037	$a_6a_3^3a_2^3$	S	1,438,751	183,248		
038	$d_4a_4^2a_4a_2^2$	S	4,421,657	501,305		
039	$a_5d_4a_3^2a_3a_1^2a_1$	S	3,948,897	463,095		
040	$a_5^2a_3^2a_1^2a_1^2a_1^2$	S	1,720,526	204,985		
041	$a_5a_4^2a_3a_3a_1$	S	4,400,394	490,201		
042	$d_4^2a_3^4$	S	546,423	63,594		
043	$a_4^3a_3^3$	S	1,051,604	129,019		
044	a_3^8	H	5,488,960†	260,864		
045	$a_5^2a_4^2a_1^2$	S	2,340,518	274,814		
046	$a_6a_4a_4a_3a_2a_1a_1$	S	4,087,798	475,079		
047	$a_5^2a_4a_3a_2^2$	S	1,834,000	214,970		
048	$a_4^4a_3^2$	S	301,557	36,564		
049	$d_5d_4a_3^4$	S	296,093	44,514		
050	$d_5a_5a_3^2a_3a_1^2a_1$	S	1,011,185	141,412		
051	$a_5^2d_4^2a_1^2$	S	318,627	44,932		
052	$a_7a_3^4a_1^4$	S	210,503	30,374		
053	$d_5a_4a_4^2a_2^2$	S	1,141,132	156,360		
054	$a_6d_4a_4^2a_2$	S	1,186,218	143,056		
055	$a_6a_5a_4a_3a_2a_1$	S	2,118,801	266,307		
056	$a_5^2d_4a_3^2a_1^2$	S	802,087	114,197		
057	$a_5a_5a_4^2a_3$	S	854,034	103,795		
058	$a_5^2d_4a_3^2a_1^2$	S	368,757	47,503		
059	$d_4^4a_1^8$	S	2,562,400	425,777		
060	$a_7a_4^2a_3^2$	S	568,181	68,406		
061	$a_6^2a_4a_3a_1^2$	S	560,384	66,786		
062	$a_6a_5a_5a_3a_2$	S	1,043,635	126,248		
063	$a_6^2a_3^2a_2^2$	S	462,316	59,317		
064	a_5^4	S	87,520	10,449		
065	$a_7a_5^2a_2^2$	S	170,548	22,325		
066	$a_7a_5a_4^2a_1$	S	304,034	38,166		
067	$a_7a_5d_4a_3a_1a_1a_1$	S	559,433	73,377		
068	$a_6d_5a_4a_4a_2$	S	620,319	78,724		
069	$d_5^2a_3^4$	S	64,133	7,984		
070	$d_5a_5^2d_4a_1^2$	S	295,716	38,802		
071	$a_5^2a_5d_4a_3a_1$	S	171,216	26,571		
072	$a_5^4a_1^4$	H	577,232†	27,736		
073	$a_8a_4^2a_3^2$	S	161,453	20,414		
074	$a_7a_6a_5a_2a_1$	S	367,327	46,932		
075	$a_6a_6a_5a_4a_1$	S	248,990	32,855		

id	roots	method	pairs	by search	via T_{18}	via T_{20}
076	$a_7^2 d_4 a_1^4$	S	52,015	7,805		
077	$d_6 a_5^2 a_3^2$	S	62,031	11,040		
078	$a_8 a_5^2 a_3$	S	103,280	13,704		
079	$d_6 a_5^2 a_3^2$	S	100,895	15,282		
080	$d_6 d_4^3 a_1^6$	S	31,122	6,099		
081	$a_7 d_5 a_5 a_3 a_1 a_1 a_1$	S	154,990	22,363		
082	$d_5 d_5 a_5^2 a_1^2$	S	90,976	14,071		
083	$a_7 d_5 d_4 a_3 a_3$	S	166,234	23,010		
084	$a_6 a_6 d_5 a_4$	S	169,401	21,966		
085	$d_5^2 a_5^2 a_1^2$	S	38,140	5,116		
086	$a_6^2 a_5^2$	S	74,583	10,603		
087	d_4^6	F	184,608†	10,992		
088	$a_8 a_6 a_5 a_2 a_1$	S	107,203	14,130		
089	$a_7^2 a_4^2$	S	49,570	6,838		
090	$a_9 a_5 d_4 a_3 a_1 a_1$	S	60,345	7,763		
091	$a_7 d_6 a_5 a_3 a_1$	S	71,608	9,666		
092	$a_8 a_6 d_5 a_2$	S	67,915	9,394		
093	$d_6 d_5 a_5^2$	S	32,952	4,944		
094	d_5^4	P				
095	$a_8 a_7 a_4 a_3$	S	144,220	20,340		
096	$a_7 d_5^2 a_3^2$	H	99,236†	4,944		
097	$a_7^2 d_4^2$	H	90,984†	5,120		
098	$a_8^2 a_3^2$	S	82,476	10,880		
099	$a_9 a_6 a_5 a_2$	S	93,802	12,782		
100	a_7^3	S	58,356	8,658		
101	$d_7 a_5^2 a_5 a_1$	S	62,300	12,014		
102	$a_9 d_5^2 a_1^2 a_1$	S	49,692	8,204		
103	$a_7 e_6 d_4 a_3^2$	S	17,228	3,172		
104	$e_6 a_6^2 a_4$	S	70,108	13,108		
105	$e_6 d_5 a_5^2 a_1^2$	H				
106	$d_6^2 d_4^2 a_1^4$	H	49,612†	3,408		
107	$a_7^2 d_6 a_1^2$	S	49,804	8,360		
108	$a_7^2 d_5 d_4$	H	40,252†	1,814		
109	$a_{10} a_6 a_5 a_1$	S	16,386	1,934		
110	$a_8 a_7^2$	S	6,920	816		
111	$a_9^2 a_2^2$	F	30,012†	2,348		
112	$d_7 a_7 d_5 a_3$	S	20,846	3,360		
113	$a_9 d_6 a_5 a_3$	S	12,184	2,142		
114	$a_9 a_8 a_5$	S	19,806	3,156		
115	$a_7^2 d_5^2$	H3	11,664†	568		
116	$a_8^2 a_7$	H	10,864†	344		
117	$d_8 d_4^4$	EP				
118	$a_9 d_7 a_5 a_1 a_1$	S	11,980	2,067		
119	$a_{11} d_5 d_4 a_3$	H	9,553†	649		
120	$a_9 a_7 d_6 a_1$	H	10,406†	537		
121	$a_9 a_7 d_6 a_1$	H	6,128†	412		
122	$d_6^2 d_6 d_4 a_1^2$	F	8,285†	553		
123	$a_{11} a_8 a_3$	F	7,618†	588		
124	$d_8 a_7^2$	F	5,136†	592		
125	$a_{11} d_6 a_5 a_1$	F	4,284†	306		

id	roots	method	pairs	by search	via T_{18}	via T_{20}
126	$a_{10}e_6a_6$	F	4,168†	384		
127	$d_7a_7e_6a_3$	H	4,020†	356		
128	$a_9e_6d_6a_1$	P				
129	$e_6^2d_5^2$	P				
130	$a_{12}a_7a_4$	H	3,988†	188		
131	$d_8d_6^2a_1^2a_1^2$	F	2,500†	354		
132	d_6^4	H3	2,416†	104		
133	$a_{13}d_6a_3a_1$	P				
134	$d_9a_7^2$	EP				
135	$a_{11}d_7d_5$	F	928†	52		
136	a_{11}^2	P				
137	$a_9e_7a_7$	F	1,208†	144		
138	$e_7d_6d_6d_4a_1$	F	1,026†	139		
139	$d_8^2d_4^2$	F	1,008†	60		
140	$a_{11}d_9a_3$	F	576†	56		
141	$a_{12}a_{11}$	F	637†	23		
142	$d_{10}d_6^2a_1^2$	EP				
143	$a_{15}a_8$	B				
144	$a_{16}a_7$	P				
145	$d_8^2d_8$	SP				
146	d_8^3	SP				
147	$a_{15}d_8$	P				
148	$a_{15}e_7a_1$	P				
149	$d_{10}e_7d_6a_1$	S	44	44		
150	$d_8e_7^2a_1^2$	P				
151	$d_{12}d_8d_4$	SP				
152	$e_8d_8^2$	SP				
153	d_{12}^2	So				
154	a_{23}	P				
155	$d_{16}d_8$	SP				
total (rows without †)			593,699,707	25,775,627	5,190,752	2,508,295

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